

Midterm 1 Comprehensive Study Guide — Stat 490

Design of Experiments · Spring 2026

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1. Foundations: Hypothesis Testing & Two-Sample Inference

1.1 Core Definitions

- Population: full set of units. Sample: subset used for inference. SRS: every unit has equal chance.
- H_0 (null): no effect/difference. H_a (alternative): effect/difference exists.
- Type I error (α): reject H_0 when true (false positive). Type II (β): fail to reject H_0 when H_a true (false negative).
- p-value: $P(\text{test stat as or more extreme} \mid H_0 \text{ true})$. Reject H_0 if p-value $< \alpha$.
- Power = $1 - \beta = P(\text{reject } H_0 \mid H_a \text{ true})$. Increases with larger n , larger effect size, larger α , smaller σ .

1.2 t-Tests: Assumptions & Theory

One-sample t : test $H_0 : \mu = \mu_0$.

- Assumptions: SRS from population; errors $X - \mu$ i.i.d; population approx normal or n large.
- Test statistic: $t = (\bar{X} - \mu_0)/(s/\sqrt{n})$; compares observed mean shift to its SE under H_0 .

Two-sample pooled t : $H_0 : \mu_1 = \mu_2$ with equal variances.

- Assumptions: two independent SRSs; both populations approx normal; $\sigma_1^2 = \sigma_2^2$ (check SD ratio in $[1/2, 2]$, F/Bartlett/Levene).
- Test stat: $t = (\bar{X}_1 - \bar{X}_2)/(s_p \sqrt{1/n_1 + 1/n_2})$, where s_p^2 pools both sample variances.

Welch (non-pooled) t : $H_0 : \mu_1 = \mu_2$ with unequal variances.

- Assumptions: two independent SRSs; normal or large n ; variance may differ but shapes similar.
- Test stat: $t = (\bar{X}_1 - \bar{X}_2)/\sqrt{s_1^2/n_1 + s_2^2/n_2}$ with Satterthwaite df.

Paired t : $H_0 : \mu_d = 0$ for differences.

- Assumptions: pairs form SRS of differences; differences d_i approx normal or n large; pairs independent.
- Test stat: $t = \bar{d}/(s_d/\sqrt{n_d})$; reduces variability by blocking at subject level.

Scenario	Test Statistic	df
One-sample	$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$	$n - 1$
Pooled two-sample	$t = (\bar{x}_1 - \bar{x}_2)/(s_p \sqrt{1/n_1 + 1/n_2})$	$n_1 + n_2 - 2$
Welch two-sample	$t = (\bar{x}_1 - \bar{x}_2)/\sqrt{s_1^2/n_1 + s_2^2/n_2}$	Satterthwaite
Paired	$t = \bar{d}/(s_d/\sqrt{n_d})$	$n_d - 1$

TI-84: STAT → TESTS → 2:T-Test (one-sample), 4:2-SampTTest (two-sample). For pooled, set Pooled:Yes. Use Stats input to enter $\bar{x}, s, n$ directly.

1.3 CIs & Assumption Checks

CI for μ : $\bar{x} \pm t_{\alpha/2,df} s/\sqrt{n}$. CI for $\mu_1 - \mu_2$: $(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,df} SE$; if CI excludes 0 $\Rightarrow$ reject H_0 .

- Normality: histogram, QQ-plot, Shapiro–Wilk test. t/F tests robust to mild non-normality for balanced moderate n .
- Equal variance (2 groups): F-test on ratio s_1^2/s_2^2 (sensitive to non-normality).
- Equal variance (k groups): Bartlett (powerful if normal, sensitive otherwise); Levene (robust, uses absolute deviations from median).

2. One-Way ANOVA (Fixed Effects, CRD)

2.1 Model & Assumptions

One-factor fixed-effects ANOVA model (balanced CRD):

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, \dots, a, \quad j = 1, \dots, n, \quad \varepsilon_{ij} \sim \text{NID}(0, \sigma^2).$$

Assumptions:

- Independence: ensured by random assignment (no time/space patterns).
- Normality: error terms ε_{ij} are normal within each group.
- Constant variance: $\text{Var}(\varepsilon_{ij}) = \sigma^2$ for all i, j .

Treatment means: $\mu_i = \mu + \tau_i$, with constraint $\sum_{i=1}^a \tau_i = 0$. Total $N = an$.

2.2 Decomposition, EMS, & ANOVA Table

Key identity: $SS_T = SS_{\text{Trt}} + SS_E$.

$$SS_T = \sum_{i,j} (y_{ij} - \bar{y}_{..})^2, \quad SS_{\text{Trt}} = \sum_i n(\bar{y}_{i.} - \bar{y}_{..})^2, \quad SS_E = \sum_{i,j} (y_{ij} - \bar{y}_{i.})^2.$$

ANOVA table:

Source	df	SS	MS	F
Treatment	$a - 1$	SS_{Trt}	$MS_{\text{Trt}} = SS_{\text{Trt}}/(a - 1)$	MS_{Trt}/MS_E
Error	$N - a$	SS_E	$MS_E = SS_E/(N - a)$	
Total	$N - 1$	SS_T		

Expected mean squares (fixed effects):

$$E[MS_E] = \sigma^2, \quad E[MS_{\text{Trt}}] = \sigma^2 + \frac{n \sum_{i=1}^a \tau_i^2}{a - 1}.$$

Under H_0 (all $\tau_i = 0$): both MS estimate σ^2 , so $F \approx 1$. Under H_a : $E[MS_{\text{Trt}}] > E[MS_E]$, so F tends to be large. Estimates: $\hat{\mu} = \bar{y}_{..}$, $\hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}$, $\hat{\mu}_i = \bar{y}_{i.}$, residuals $e_{ij} = y_{ij} - \bar{y}_{i.}$. Coefficient of determination: $R^2 = SS_{\text{Trt}}/SS_T$.

2.3 Regression Equivalence

ANOVA with a levels is equivalent to regression with $a - 1$ dummy variables. With reference level 1:

$$y_{ij} = \beta_0 + \beta_2 x_{2,ij} + \cdots + \beta_a x_{a,ij} + \varepsilon_{ij},$$

where β_0 is mean of level 1 and β_i are differences from level 1. ANOVA F for factor = regression F for all dummies.

2.4 CIs from ANOVA

Using MS_E from ANOVA and error df = $N - a$:

- CI for μ_i : $\bar{y}_{i.} \pm t_{\alpha/2, N-a} \sqrt{MS_E/n}$.
- CI for $\mu_i - \mu_j$: $(\bar{y}_{i.} - \bar{y}_{j.}) \pm t_{\alpha/2, N-a} \sqrt{2MS_E/n}$ (balanced).

2.5 Worked Example: Dog Food Shelf Height

3 shelf heights (Knee, Eye, Waist), $n = 8$ stores per group, $N = 24$ total.

	Knee	Eye	Waist
$\bar{y}_{i.}$	81.0	84.625	90.875

$\bar{y}_{..} = 85.5$, $SS_{\text{Trt}} = 399.2$, $SS_E = 288.7$, $MS_E = 13.75$. Then $F = 199.62/13.75 = 14.52$ with df (2, 21), p-value ≈ 0.00011 ; strong evidence of unequal means.

95% CI for μ_{Knee} : $81 \pm 2.08 \sqrt{13.75/8} = 81 \pm 2.73 = (78.27, 83.73)$.

TI-84: `Fcdf(14.52,1E99,2,21)` gives p-value. `invT(0.025,21)` gives $t_{0.025,21}$. Use **1-Var Stats** on each group list to get $\bar{x}, s^2, n$.

3. Multiple Comparisons

3.1 Familywise Error & Concepts

With K tests at level α , FW (experiment-wise) Type I error $\approx 1 - (1 - \alpha)^K$. For a means, $K = \binom{a}{2}$ pairwise tests. For $a = 4$ and $\alpha = 0.05$, FW error ≈ 0.26 .

Family definition matters: could be “all pairwise comparisons”, a set of planned contrasts, or all contrasts considered after seeing data.

3.2 Post-Hoc Procedures: Assumptions & Use

Method	Key Idea / Formula	When to Use	FWER?
Fisher LSD	LSD = $t_{\alpha/2, df_E} \sqrt{2MS_E/n}$; pair i, j sig if $ \bar{y}_{i.} - \bar{y}_{j.} > \text{LSD}$.	After sig ANOVA F ; few groups; liberal.	No
Bonferroni	Each test at α/K or CIs at $1 - \alpha/K$.	Any small family of planned tests.	Yes
Tukey HSD	HSD = $q_{\alpha, a, df_E} \sqrt{MS_E/n}$.	All pairwise comparisons; equal variances; near-balanced.	Yes
Dunnett	Critical values for trt vs control differences only.	All trts vs single control.	Yes
Scheffé	Reject if $\frac{ \hat{\Gamma} /SE}{\sqrt{(a-1)F_{\alpha, a-1, df_E}}} >$	Any (even post-hoc) contrast.	Yes

Assumptions: all inherit ANOVA assumptions (normal, equal variance, independence). Tukey/Dunnett additionally assume equal MS_E across groups; Scheffé valid for arbitrary contrasts, even selected after seeing data.

3.3 Dog Food Pairwise Comparisons

From §2.5: $a = 3$, $n = 8$, $MS_E = 13.75$, $df_E = 21$. $SE = \sqrt{2MS_E/n} = 1.854$.

Pair	$ \bar{y}_{i.} - \bar{y}_{j.} $	LSD	Tukey HSD	Bonf. thresh
W–K	9.875	3.86	4.67	4.86
W–E	6.250	3.86	4.67	4.86
K–E	3.625	3.86	4.67	4.86

All methods find W differs from K and E; K vs E not significant.

3.4 Contrasts & Orthogonality

A contrast is $\Gamma = \sum_{i=1}^a c_i \mu_i$ with $\sum c_i = 0$. Estimate $\hat{\Gamma} = \sum c_i \bar{y}_{i.}$; $SE(\hat{\Gamma}) = \sqrt{MS_E \sum c_i^2/n}$ (balanced). CI: $\hat{\Gamma} \pm t_{\alpha/2, df_E} SE(\hat{\Gamma})$. Example: control vs average of two treatments: $C = (-2, 1, 1)$; protein effect in 2×2 : $C = (0.5, 0.5, -0.5, -0.5)$.

Two contrasts C, D are orthogonal if $\sum c_i d_i = 0$ (balanced). There are at most $a - 1$ mutually orthogonal contrasts; their SS add to SS_{Trt} (each 1 df).

3.5 Data Snooping & Scheffé

Choosing contrasts after looking at data inflates Type I error (implicit multiple testing). Scheffé’s method controls FW error for *all* contrasts at once; more conservative than Tukey for pairwise comparisons but valid under data snooping.

4. Model Adequacy & Transformations

4.1 Residual Diagnostics

Residuals $e_{ij} = y_{ij} - \bar{y}_{i.}$ summarize unexplained variation. Checks:

- Normality: QQ-plot; Shapiro–Wilk test.
- Constant variance: residuals vs fitted; look for funnels. Bartlett/Levene tests.
- Outliers: studentized residual $|e^*| > 3$ suspicious.
- Independence: residuals vs run order for time trends.

4.2 Variance-Stabilizing Transformations

If $\text{Var}(Y) = g(\mu)$, find h with $h'(\mu) \propto 1/\sqrt{g(\mu)}$ so $\text{Var}(h(Y))$ is roughly constant (delta method). Common cases:

Distribution	$\text{Var}(Y)$	Transform	Example
Poisson	μ	$h(y) = \sqrt{y}$	Flaw counts
Binomial	$n\mu(1 - \mu)$	$h(y) = \sin^{-1} \sqrt{y}$	Proportions
Lognormal-like	$\propto \mu^2$	$h(y) = \ln y$	Times, prices

Decision rule: if variance grows with mean, try log or square root; if data are proportions near 0/1, try arcsin-sqrt. If transformations fail and assumptions still poor, use non-parametric tests.

TI-84: store raw data in L1, set L2 = $\sqrt{\text{L1}}$ or L2 = $\ln(\text{L1})$, then run 1-Var Stats L2.

5. Random Effects Model

5.1 Fixed vs Random Effects

Fixed effects: treatment levels are the *only* ones of interest; inference about those specific means. Random effects: levels are randomly sampled; inference about variance among levels and the population of levels.

Examples: specific 3 labs vs 3 labs sampled from all labs; specific therapists vs random sample of therapists.

5.2 Model, Assumptions, EMS

Random one-way ANOVA:

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad \tau_i \sim N(0, \sigma_\tau^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2), \text{ independent.}$$

Assumptions:

- Levels i are random sample from population; τ_i i.i.d with mean 0, variance σ_τ^2 .
- Errors ε_{ij} i.i.d $N(0, \sigma^2)$ and independent of τ_i .

Variance structure:

- $\text{Var}(Y_{ij}) = \sigma_\tau^2 + \sigma^2$.
- $\text{Cov}(Y_{ij}, Y_{ij'}) = \sigma_\tau^2$ (same level). Different levels: covariance 0.

Expected mean squares:

$$E[MS_E] = \sigma^2, \quad E[MS_{\text{Trt}}] = \sigma^2 + n\sigma_\tau^2.$$

F-test remains $F = MS_{\text{Trt}}/MS_E$ to test $H_0 : \sigma_\tau^2 = 0$.

5.3 Variance Components & ICC

Estimators:

- $\hat{\sigma}^2 = MS_E$.
- $\hat{\sigma}_\tau^2 = (MS_{\text{Trt}} - MS_E)/n$ (may be negative if true variance small; then set to 0 in practice).

Intraclass correlation:

$$\rho = \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma^2} = \text{Corr}(Y_{ij}, Y_{ij'}).$$

Interpretation: fraction of total variance explained by group identity. If $\rho \approx 0$, grouping has little effect; ρ near 1 means within-group units nearly identical.

5.4 Example: Sires & Calf Weights

5 sires (random), 8 calves each: $N = 40$. From ANOVA: $MS_{\text{Trt}} = 1397.8$, $MS_E = 463.8$, $F = 3.014$, $p \approx 0.031$.

$\hat{\sigma}^2 = 463.8$, $\hat{\sigma}_\tau^2 = (1397.8 - 463.8)/8 = 116.75$, $\text{ICC} = 116.75/(116.75 + 463.8) = 0.201$.

CI for overall mean (random): $82.55 \pm t_{0.025,35} \sqrt{MS_{\text{Trt}}/N} \approx (70.5, 94.6)$; fixed-effects CI using MS_E would be narrower.

TI-84: Fcdf(3.014,1E99,4,35) for F p-value. For CI on σ^2 , use χ^2 distribution with $\text{df} = N - a$.

6. RCBD (Blocking)

6.1 Motivation, Model, Assumptions

Blocking: group experimental units into homogeneous blocks for nuisance factors (location, day, operator) and randomize treatments within each block.

RCBD model (fixed blocks):

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad i = 1, \dots, a, \quad j = 1, \dots, b.$$

Assumptions:

- Errors ε_{ij} independent, $N(0, \sigma^2)$.
- Additive model: no treatment $\times$ block interaction; difference between any two treatments same across all blocks.

6.2 ANOVA Table & df

Source	df	SS	F
Treatment	$a - 1$	SS_{Trt}	MS_{Trt}/MS_E
Block	$b - 1$	SS_{Block}	MS_{Block}/MS_E
Error	$(a - 1)(b - 1)$	SS_E	
Total	$ab - 1$	SS_T	

Error df: $(a - 1)(b - 1)$. For $a = 4, b = 3$, $df_E = (4 - 1)(3 - 1) = 6$.

6.3 Why Blocking Helps

Apple orchard example (5 treatments, 6 locations):

Design	MS_E	Conclusion
CRD	52,938	No trt diff; high variance
RCBD	34,455	Same conclusion, but 35% lower MS_E

Even if the overall conclusion is unchanged here, reduced MS_E gives more power in other settings.

6.4 Mixed RCBD: Random Blocks

Random blocks model: $Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$ with $\beta_j \sim N(0, \sigma_\beta^2)$ i.i.d, treatments fixed.

- ANOVA table and dfs unchanged; treatment F-test still MS_{Trt}/MS_E .
- Block variance component: $\hat{\sigma}_\beta^2 = (MS_{\text{Block}} - MS_E)/a$.
- Focus of inference: treatment means/effects; blocks model nuisance variability.

If interaction between blocks and treatments is real but not modeled, interaction gets absorbed into SS_E , inflating MS_E and reducing power.

6.5 Multiple Comparisons in RCBD

Post-hoc tests for treatment means use same logic as CRD but with:

- Error df = $(a - 1)(b - 1)$.
- Effective “replicates” per treatment = b (one per block).
- $SE(\bar{y}_{i.} - \bar{y}_{j.}) = \sqrt{2MS_E/b}$.

7. Power & Sample Size

7.1 Concepts

Power: $P(\text{reject } H_0 \mid H_a \text{ true})$. Depends on α, n , effect size, and σ^2 .

- Larger power (e.g., 0.9 vs 0.8) $\Rightarrow$ need larger n .
- Smaller α (e.g., 0.01 vs 0.05) $\Rightarrow$ larger n .
- Larger $\sigma^2 \Rightarrow$ larger n .
- Larger effect size $\Rightarrow$ smaller n .

7.2 Cohen’s f for One-Way ANOVA

Effect size:

$$f = \sqrt{\frac{\frac{1}{a} \sum_{i=1}^a (\mu_i - \bar{\mu})^2}{\sigma^2}}, \quad \bar{\mu} = \frac{1}{a} \sum \mu_i.$$

Benchmarks: $f = 0.10$ (small), 0.25 (medium), 0.40 (large). Required n grows like $1/f^2$.

7.3 Worked Example

Means (50, 60, 50, 60), $a = 4, \sigma^2 = 25$. $\bar{\mu} = 55$. Between variance = 25, so $f = 1.0$ (very large). Standard power formulas give $n \approx 3$ per group for 90% power at $\alpha = 0.05$.

If $\sigma = 6, f = \sqrt{25/36} = 0.833$; if $\sigma = 7, f = \sqrt{25/49} = 0.714$. As σ doubles, f halves and n roughly quadruples.

7.4 CI-Based Sample Size

For CI on $\mu_i - \mu_j$ with half-width ME at confidence $1 - \alpha$:

$$ME = t_{\alpha/2, df} \sqrt{2MS_E/n}.$$

Solve for n , iterating because df depends on n . For multiple intervals, may apply Bonferroni (replace α by α/K).

8. Non-Parametric Alternatives

8.1 When to Use

Use when normality/variance assumptions badly violated and transformations fail. Trade lower power under normality for fewer assumptions.

Parametric	Non-parametric	Key Assumption	Target
Paired t	Sign test	None	Median diff $\neq 0$
Paired t	Wilcoxon signed-rank	Symmetric diff dist	Location shift
Two-sample t	Mann–Whitney	Similar shapes	Location shift / $P(X > Y)$
ANOVA F	Kruskal–Wallis	Continuous data	Equal distributions

8.2 Wilcoxon Signed-Rank

For paired data or one-sample median test. Steps: compute $d_i = x_i - m_0$, drop zeros; rank $|d_i|$; sign ranks; sum positive ranks W^+ . Under H_0 , $E(W^+) = n(n+1)/4$, $\text{Var}(W^+) = n(n+1)(2n+1)/24$; use normal approx for moderate n . Assumption: distribution of differences is symmetric; no normality needed.

8.3 Mann-Whitney Rank-Sum

Two independent groups sizes m, n . Rank pooled data; let W_1 be rank sum for group 1. Convert to $U_1 = W_1 - m(m+1)/2$. Under H_0 , $E(U_1) = mn/2$, $\text{Var}(U_1) = mn(m+n+1)/12$. Assumptions: independent samples, continuous distributions, similar shapes across groups; tests location shift.

8.4 Kruskal-Wallis H

Rank all N observations; let T_i be rank sum in group i of size n_i .

$$H = \frac{12}{N(N+1)} \sum_{i=1}^a \frac{T_i^2}{n_i} - 3(N+1),$$

approx χ^2_{a-1} under H_0 . Assumptions: independent samples, continuous distributions. Under true normality with equal variances, ANOVA F more powerful; under heavy skew/outliers, K-W can be preferred.

TI-84: for χ^2 p-value use `chi2cdf(H,1E99,a-1)`; for normal approximations use `normalcdf`.

9. Experimental Design Principles

9.1 Three Principles

- Randomization: randomly assign treatments to units; protects against bias, justifies independence.
- Replication: multiple units per treatment; estimates error variance and increases precision.
- Blocking: group similar units into blocks; removes block-to-block variability from error term.

9.2 Design Comparison

Design	Nuisance factor	Model	Error df
CRD	None	$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$	$N - a$
RCBD	1 block	$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$	$(a - 1)(b - 1)$
Latin Sq	2 blocks	$y_{ijk} = \mu + \tau_j + \rho_i + \gamma_k + \varepsilon$	$(p - 1)(p - 2)$

9.3 Quick Concept Checks

- ANOVA treats quantitative and qualitative factors the same in SS computations. (True)
- Fisher LSD is extremely conservative. (False; it is liberal.)
- Tukey HSD compares all treatments to a control. (False; Dunnett does.)
- MS_E estimates the random error variance σ^2 . (True.)
- In mixed RCBD with random blocks, inference focuses on treatments, not blocks. (True.)
- Power of Kruskal-Wallis is less than F-test under normality. (True.)

10. Common Distributions & TI-84 Reference

Distribution	Mean	Variance	Use
$N(\mu, \sigma^2)$	μ	σ^2	Errors, CLT
t_ν	0	$\nu/(\nu - 2)$ ($\nu > 2$)	Small-sample means
F_{d_1, d_2}	$d_2/(d_2 - 2)$ ($d_2 > 2$)	Variance ratios, ANOVA	
χ^2_ν	ν	2ν	Variance CIs, K-W
Poisson(λ)	λ	λ	Counts; $\sqrt{Y}$ transform
Binomial(n, p)	np	$np(1 - p)$	Proportions; $\arcsin\sqrt{Y}$

TI-84 distributions (2nd→DISTR): normal: `normalcdf`, `invNorm`; t: `tcdf`, `invT`; F: `Fcdf`; χ^2 : `chi2cdf`. Example: upper-tail p-value for $F=14.52$, $df=(2,21)$: `Fcdf(14.52,1E99,2,21)`.

Use margins and blank areas for your own textbook examples and quick formulas.